

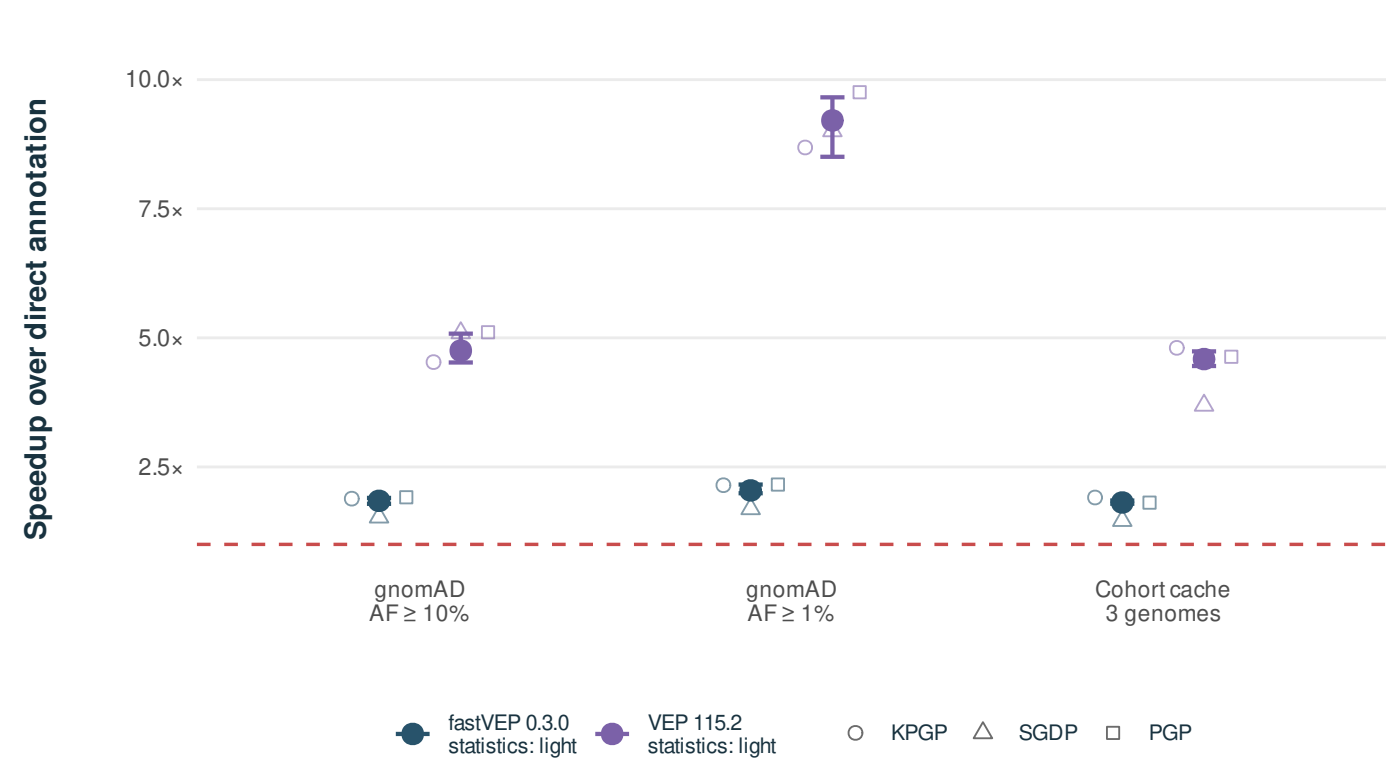
VCFcache reduces runtime for both VEP and fastVEP

Exactly paired genomes, variants, cache blueprints and light-statistics timing mode

A

Caching helps both annotators

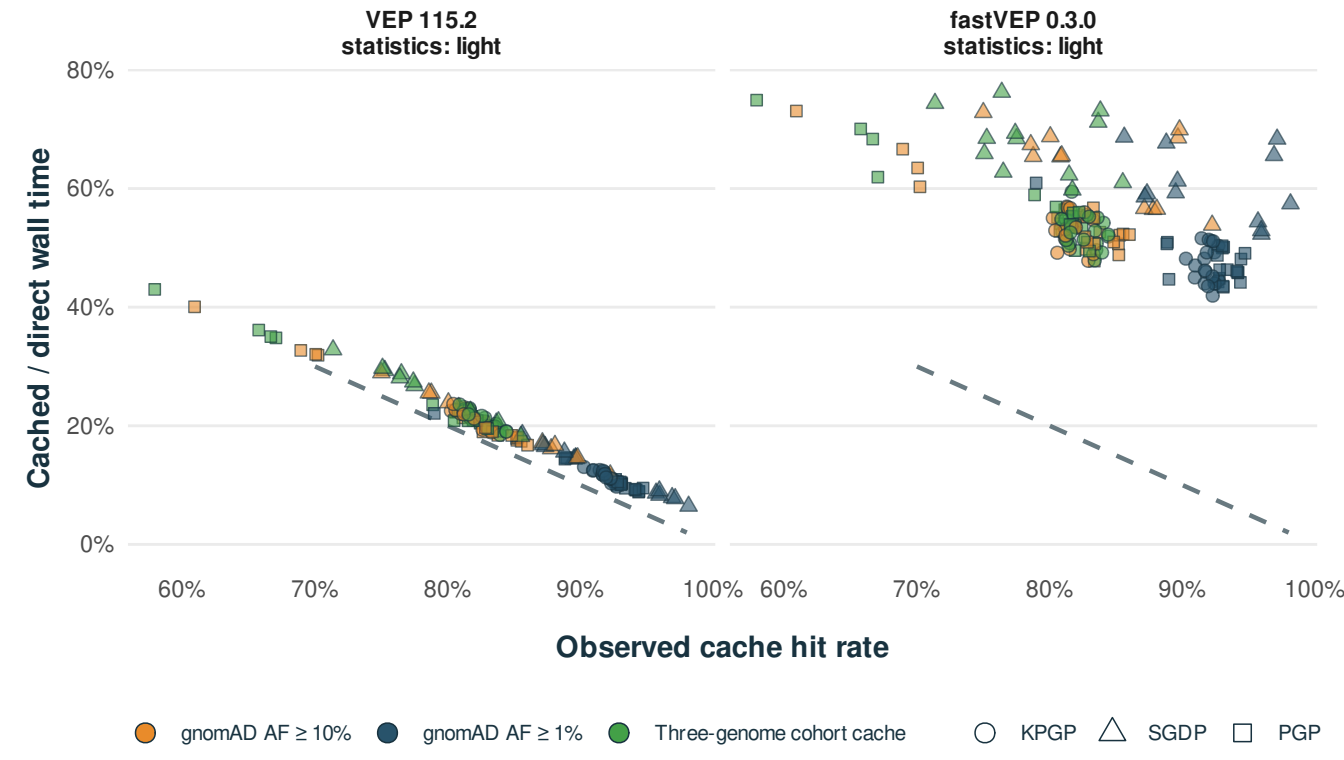
Large points: stratified median with 95% bootstrap interval; small symbols: cohort medians



B

The same hits remove more work from the slower pipeline

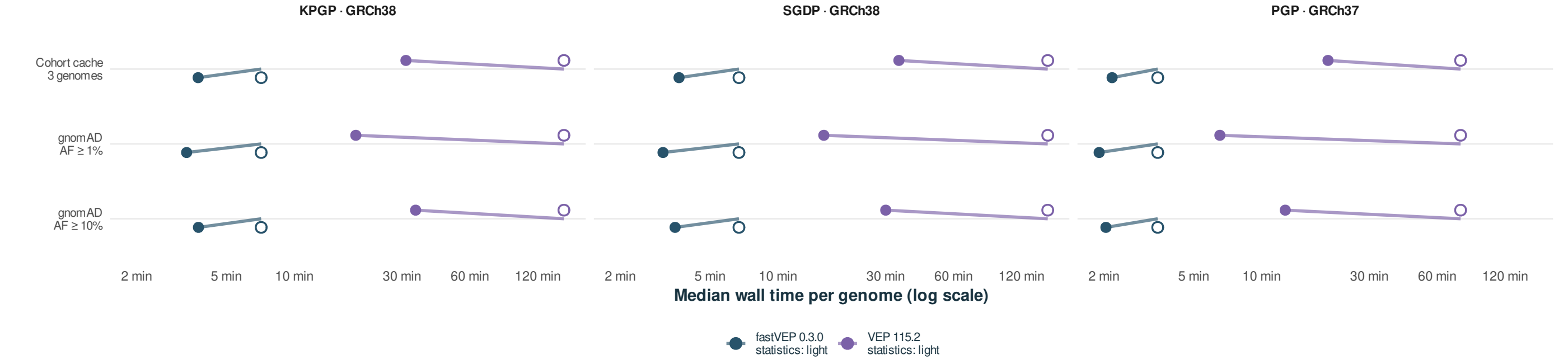
Dashed line: zero-overhead miss-fraction rule; axes are identical



C

Absolute wall time spans minutes to hours

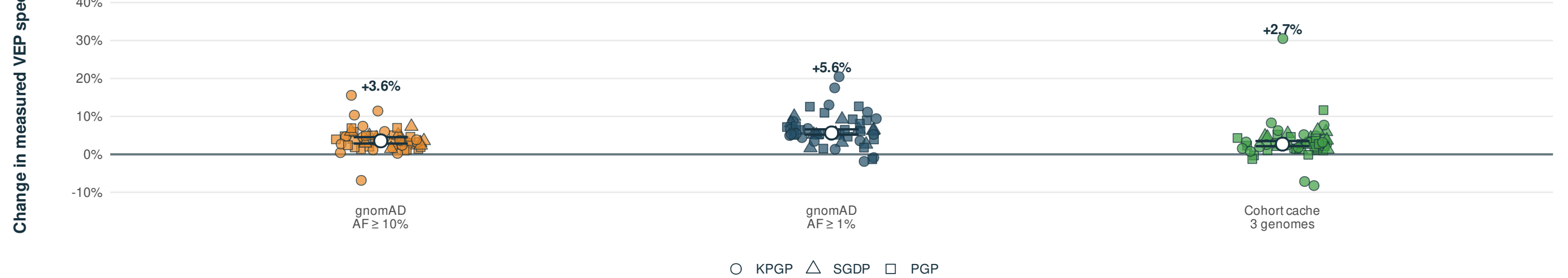
Open point: direct; filled point: cached; line: time returned per genome



D

The full-cohort rerun confirms that the legacy rescan was conservative

All 52 genomes paired by sample and strategy; points compare light/full speedup, labels show medians



VEP 115.2 and fastVEP 0.3.0 were both timed with VCFcache --statistics light across all 52 genomes.
Panel D pairs the new VEP light measurements with the historical full-rescan values; no observation was rescaled.
Each genome/tool/strategy combination was run once; stratified bootstrap intervals treat genomes as inferential units.